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Summary

AI engineer who owns products end to end for real businesses. At Lakestone AI I am embedded with each client, turning their back-office rules into shipped software: extraction pipelines with judge models behind them, agents that act within limits a person sets, assistants that answer from company data without leaking it, and the ERP integrations underneath. Eight years of full-stack work across TypeScript, Node.js, React, .NET and Python, the last few focused on LLM pipelines, evals and agent tooling. Claude Code is my primary development environment; the engineering is in how agents are briefed, constrained and checked.

Experience

Forward Deployed Engineer & Lead AI Engineer — Lakestone AI, Perth

August 2026 – present

AI back-office platform for a construction company (lead engineer, live in production). Supplier invoices were re-keyed into SimPro by hand and ~2,000 suppliers were onboarded over email.

- Designed the invoice agent: Gemini extraction into Zod-validated schemas, confidence and drift gates, straight-through to a SimPro vendor order and receipt, escalation with the reason. A GST TaxAdjustment lands totals on the printed figure, proven by a live read-back check; kept the human approval gate until the judge layer earned the right to bypass it.
- Built the LLM-as-judge layer: JEV scores every classification against the source, CDM arbitrates between candidate extractions; below-threshold results go to a person. Judges are versioned prompts in Langfuse with promptfoo suites, benchmarked like any model change.
- Supplier agent: public tokenised onboarding with ABN/ABR, AusPayNet BSB and certificate checks that fail open and never silently, banking-change fraud screening with ABA pre-flight, SimPro vendor create-or-link. 134/137 fields extracted correctly on staff-uploaded forms.
- Inspection agent: tokenised driver requests, mail and SharePoint ingest, three-run Gemini consensus extraction, SharePoint write-back. Redesigned the vehicle mirror with per-root delta cursors after a stale cursor let 29 nightly syncs report success while doing nothing.
- Chat assistant over invoices, suppliers, vehicles and a 7,367-document corpus (Postgres FTS first, pgvector hybrid second, per-tree permissions). Moved its SQL tool from a parsing allowlist to a dedicated NOLOGIN Postgres role via PostgREST JWT role-switching; exploit tests run in CI against a replayed database.
- 276 migrations, 96 RLS policies, Vitest coverage floors, Playwright, pgTAP; staging-to-main release model; SimPro mirror fed by 34+ webhook types and reconciled daily against the live API.

Mail, property and agreement platform for a commercial property agency (live 14 Oct 2026). Microsoft 365 mail synced and classified by Gemini (judged by JEV); every proposal and reply approved by a person.

- Owned client research, the Cirrus8 integration (proved a read API the roadmap assumed was CSV-only; built a mirror whose freshness is a health check), the MAA workshop and generator, schema, access control and the production go-live.
- Access control in one day: roles, permissions, global admins, access log, restrictive write policies, pgTAP permission matrices; 201 RLS policies covering every application table; separate database logins per service; Entra sign-in.
- Shipped a release that had not built for three weeks: three weeks of accumulated changes merged as one release, 94 migrations replayed, tier guard changed with recorded reasons, 42 stale env overrides removed, readiness measured.

Headless e-signature service and PDF toolkit (sole author, v1.2, in production). Built in six days to replace a per-envelope SaaS subscription.

- Email-OTP signers, field rendering, CAdES detached seals with our own CMS/DER, byte-range and X.509 chain code, HMAC-signed webhooks; sealed PDFs verify against the CA by a script with its own tests.
- 40+ PDF tools in TypeScript (organise, fill, redact with audit trail, protect, sanitise, compare, optimise, convert, PDF/A, OCR); per-project feature modules with requireFeature on every route and job; one CLI an agent can drive; loopback-only development proven by egress guards; licence hygiene as a CI gate.

Agent delivery pipeline and skills library.

- Designed the team's standard agent workflow: brief (options with explicit weaknesses) → one builder agent per task in its own worktree → ship (nine stages to an evidence-backed PR with adversarial review) → babysit (classifies CI failures, never weakens a check), with a model-routing table per step.

- Defined the five-role agent set (builder, reviewer, inspector, educator, mechanical) with write-blocking hooks; wrote 76 incident-born project skills; part of a three-person library of 108 skills with publish, audit and sync tooling.

Systems Developer — Ear Science Institute Australia, Perth

2025 – August 2026 · medical research not-for-profit

- Built the Finance Sync App keeping the Manage 9 practice-management system and NetSuite aligned, replacing manual reconciliation.
- Java and Python microservices on AWS Lambda and ECS, provisioned with Terraform, deployed through GitLab CI/CD; NetSuite SuiteScript 2.1 customisation; Windows Server, Hyper-V and Linux infrastructure.
- MCP NetSuite–Ollama bridge: an MCP server fronting a local Ollama model so marketing could query the ERP in plain English with inference on-premises; v2 added LangGraph multi-step reasoning. Ran in production.
- n8n marketing-research workflows that cut manual collection work by about 70%.

Graduate Systems Developer → Solutions Architect — GlobalCharge Limited, UK

Payments company; merchants included Badoo and Bumble

- Co-developed a merchant payment portal on AWS integrated with multiple banks; built the gateway's standing-order service.
- Implemented OAuth, JWT, AES-256 encryption and RBAC to data-protection requirements; applied OWASP practices across the codebase; ran MongoDB and PostgreSQL in production.

Skills

AI	Gemini, Claude, DeepSeek, Vercel AI SDK, LangChain/LangGraph, MCP, Ollama, Langfuse, promptfoo, LLM-as-judge, RAG with pgvector + Postgres FTS, agent design, Claude Code skills and hooks
Languages	TypeScript, JavaScript, Python, Java, C#, SQL
Backend	Bun, Hono, Node.js, Django/DRF, .NET, FastAPI, PostgreSQL (Supabase, RLS, pgTAP, postgres.js), MongoDB, Redis
Frontend	React 19, Vite, TanStack Router/Query/Table, Tailwind v4, shadcn/ui, Next.js, Angular
Jobs & infra	Trigger.dev, Vercel Workflow SDK, Celery, AWS (Lambda, ECS, S3), Vercel, Docker Compose, Caddy, Terraform, Doppler, GitHub Actions, GitLab CI
Integrations	NetSuite SuiteScript 2.1, SimPro API, Microsoft Graph (SharePoint, mail), Entra ID, Australian Business Register, AusPayNet, Cirrus8, Salesforce, MYOB, Slack, n8n
Documents & crypto	PDF internals (xref, XMP, PDF/A, fonts, ICC), CMS/DER, CAdES, X.509, OCR, HMAC webhooks, OTP flows

How I work

- **A second model checks the first.** Nothing an extractor or classifier produces reaches a client system without a judge pass.
- **Fail visible.** AI checks fail open for the user and never silently; background tasks abort with a named error.
- **Verify, don't trust.** Tests drive real HTTP surfaces and assert on data; delegated agent work is re-run and read before it is accepted.
- **Client systems stay the source of truth.** Mirror, reconcile and write back; never fork their data.

Most production work is in private client repositories; I am happy to walk through it in a call. Open to forward deployed, AI engineering and senior full-stack roles, remote or hybrid.